Identify a project based on the quantum computing course syllabus. You can choose a batch size of upto 3 students. Individual members of the team should present their contributions in detail. Some sample projects are given below. It is recommended to find your own real-life projects.

1) A program that converts matrix/vector representations into Dirac notation and vice versa.
2) Write code to implement QFT and compare with classical FFT for small datasets.
3) Factorize integers using Shor's algorithm in Qiskit.
4) Evaluate Qiskit, Cirq, and PyQuil by implementing the same algorithm and comparing performance.
5) Apply quantum Fourier transform to small grayscale images and compare with classical FFT.
6) Build a **symbolic calculator** for Dirac notation
7) Use **PennyLane or Qiskit** to solve a simple optimization problem
8) Implement a **quantum neural network (QNN)** for a simple classification task
9) Encode sentences into **quantum states** and perform simple NLP tasks (e.g., sentiment analysis).
10) Design quantum circuits to solve puzzles.